A SURJECTIVE ENDOMORPHISM OF A NOETHERIAN RING IS AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Noetherianity as the ascending chain condition on ideals. Supplies the contradiction at the end of the counted proof (line 6446). Forbids the strictly increasing chain of kernels produced from a non-injective endomorphism.
- Iterated kernels of a non-injective endomorphism. Built inside the counted proof at lines 6441-6445. Produces the chain of ideals whose strict increase contradicts Noetherianity.

Lemma 0.1. *Any surjective endomorphism of a Noetherian ring is an isomorphism.*

Proof. If $f : R \to R$ were such an endomorphism but not injective, then

$$\mathrm{Ker}(f) \subset \mathrm{Ker}(f \circ f) \subset \mathrm{Ker}(f \circ f \circ f) \subset \ldots$$

would be a strictly increasing chain of ideals. □

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